CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm. Examination

University Theory Examination October 2017

PH324 Pharmaceutical Analysis - II

Date: 03.10.2017, Tuesday **Time: 10.00 a.m. to 01.00 p.m.** **Maximum Marks: 80**

Instructions:

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat labeled sketches wherever necessary.**

SECTION – I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. According to Ohm's Law, the strength of current flowing through a conductor is directly proportional to the potential difference (E) applied across the conductor and inversely proportional to the _______________ of the conductor.

[A] Conductance
[B] Resistance
[C] Specific Conductance
[D] Electrial conductivity

2. Saturated Calomel Electrode is prepared using ____________ which is commonly known as Calomel

[A] Mercury
[B] Potassium chloride
[C] Mercurous chloride
[D] Silver chloride

3. Which of the following is used as an Indicator electrode.

[A] Mercury/mercurous sulphate electrode
[B] Calomel electrode
[C] Silver/Silver chloride electrode
[D] Quinhydrone electrode

4. _____________ is used for measurement of pH.

[A] Glass electode
[B] Antimony electrode
[C] Saturated calomel electrode
[D] Silver/silver chloride electrode

5. Polarimetry is used for analysis of __________________ compounds.

[A] Optically active

[B] Ionic
[C] Oxidizing
[D] Acidic

6. ______________ is used as light source in polarimetry.

[A] Sodium vapour lamp
[B] Deuterium lamp
[C] Tungston lamp
[D] Mercury lamp

7. _____________ is used as stationary phase in GSC analysis.

[A] Solid
[B] Liquid
[C] Gas
[D] Plasma

8. _____________ is used as GC detector in moisture content analysis.

[A] TCD
[B] FID
[C] ECD
[D] FPD

9. Gas Chromatography is used for

[A] Non-volatile compounds
[B] Volatile compounds
[C] Thermolabile compounds
[D] Highly reactive compounds

10. Term A in van Deemter equation is

[A] Eddy diffusion
[B] Longitudinal diffusion
[C] Mass transfer
[D] Rate constant

11. ________________ is most commonly used detector in HPLC analysis.

[A] Fluorescence detector
[B] Refractive index detector
[C] Mass detector
[D] UV/Visible absorbance detector

12. ________________ is non-absorbance detector for HPLC system.

[A] Fluorescence detector
[B] Refractive index detector
[C] PDA detector
[D] UV/Visible absorbance detector

13. ________________ is non-polar stationary phase and most commonly used in Reversed-phase HPLC.

[A] Silica
[B] C_{18}
[C] C_4
[D] Cyano

14. In __________ exchange chromatography, positively charged molecules are attracted to a negatively charged solid support.

[A] Cation

[B] Anion
[C] Zwitterion
[D] Common ion

15. Size Exclusion Chromatography is also known as

[A] Gel filtration chromatography
[B] Affinity chromatography
[C] Ion-exchange chromatography
[D] Paper chromatography

16. HPTLC tends to ______________________.

[A] High Pressure Liquid Chromatography
[B] High Pressure Thin Layer Chromatography
[C] High Performance Liquid Chromatography
[D] High Performance Thin Layer Chromatography

17. Rf value ranges from

[A] 0 to 1
[B] 0 to 10
[C] 1 to 100
[D] 1 to 14

18. ___________ is used as spraying reagent to detect the spots of amino acids in Paper Chromatography.

[A] Iodine
[B] Dragendorff's reagent
[C] Ninhydring reagent
[D] Sulphuric acid

19. __________ is an most commonly used adsorbent in TLC.

[A] Resins
[B] Agarose
[C] Cellulose
[D] Silica gel

20. _____________ chromatography is employed for purification of sample mixture.

[A] Column
[B] Thin Layer
[C] Paper
[D] HPTLC

SECTION – II

Q 2	Attempt any TWO of the following	
A	Classify instrumental methods of analysis. Write their advantages over non-instrumental methods of analysis.	**05**
B	Explain the principle of extraction. What are the different types of extraction techniques.	**05**
C	Explain the terms: Adsorption, Partitioning, Resolution, HETP, Retention time.	**05**
Q 3	Attempt any TWO of the following	
A	Describe various techniques for packing of column with stationary phase.	**05**
B	Describe various methods for chromatographic development in Paper Chromatography.	**05**
C	Discuss applications of Thin Layer Chromatography.	**05**

SECTION – III

Q 4	Attempt any **FOUR** of the following	
A	Differentiate HPLC and HPTLC.	**05**
B	Discuss stationary phase used in partition and ion exchange chromatography.	**05**
C	Write a brief note on application of column chromatography.	**05**
D	Enumerate detectors used in HPLC. Write their ideal requirement.	**05**
E	Write short note on Affinity chromatography.	**05**
F	Draw a labeled diagram of HPLC system. Write functions of each component.	**05**
Q 5	Attempt any **FOUR** of the following	
A	Draw a labeled diagram of Polarimeter. Enlist its applications in Pharmacy	**05**
B	Write a note on different types of conductometric titrations.	**05**
C	Discuss design and working principle of ANY ONE detector used in GC.	**05**
D	Write applications of Potentiometry.	**05**
E	Draw the schematic diagram of GC. Describe columns and column packing materials used in GC.	**05**
F	Explain the principle of Polarography.	**05**